DO NOW, AO1 REVISION

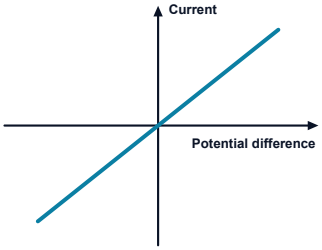
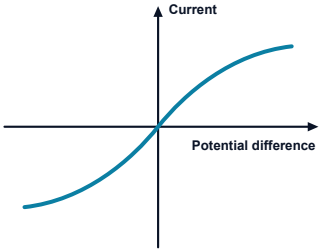
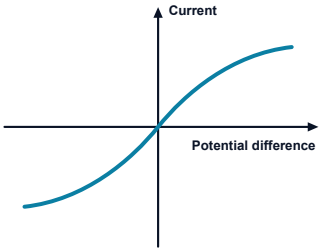
Current, resistance and potential difference

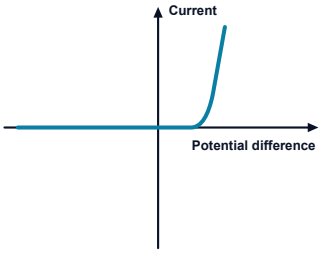
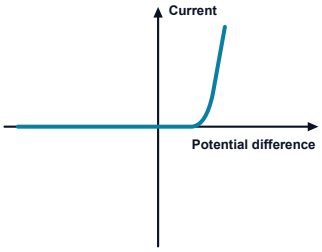
Electricity
Paper 1, Combined and Separate
24 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 12 starred (★) questions first.

	QUESTION	ANSWER	✓
1	**What is an electrical conductor?**	A material that allows charge to flow through it.	☐
2	**Name the charged subatomic particles that can flow in metals.**	Electrons	☐
3 ★	**Define electric current.**	The rate of flow of electrical charge.	☐
4 ★	**State the standard units of charge, current and time.**	Charge: coulomb (C) Current: ampere (A) Time: second (s)	☐
5 ★	**Describe the current at different points in a single closed loop.**	It is the same at every point.	☐
6	**Name the two variables that the current through a component depends on.**	The potential difference across it and its resistance	☐
7	**The potential difference across a fixed resistor is increased. Describe the effect on the current.**	The current increases.	☐
8	**The resistance of a component is increased at a constant potential difference. Describe the effect on the current.**	The current decreases.	☐
9 ★	**State the standard units of potential difference and resistance.**	Potential difference: volt (V) Resistance: ohm (Ω)	☐
10	**Name the type of component whose resistance stays constant as the current changes.**	An ohmic conductor (at constant temperature)	☐

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	QUESTION	ANSWER	✓
11 ★	**Describe how current depends on potential difference for an ohmic conductor.**	Current is directly proportional to potential difference.	☐
12	**Name the type of component that gives this I–V graph.** 	An ohmic conductor (e.g. a fixed resistor)	☐
13 ★	**Name the component that gives this I–V graph.** 	A filament lamp	☐
14	**This is the I–V graph for a filament lamp. Describe how its resistance changes as the potential difference increases.** 	The resistance increases.	☐
15 ★	**Explain why the resistance of a filament lamp increases with current.**	A greater current makes the filament hotter, and its resistance increases with temperature.	☐

	QUESTION	ANSWER	✓
16 ★	**Name the component that gives this I–V graph.** 	A diode	☐
17 ★	**This is the I–V graph for a diode. Describe its resistance when the potential difference is reversed.** 	Very high (almost no current flows).	☐
18 ★	**Describe how the resistance of a thermistor varies with temperature.**	As the temperature increases, the resistance decreases.	☐
19 ★	**Describe how the resistance of an LDR varies with light intensity.**	As the light intensity increases, the resistance decreases.	☐
20	**Give one use of a thermistor and one use of an LDR.**	Thermistor: thermostat LDR: lights that switch on when it gets dark	☐
21	**State the two things needed for electrical charge to flow in a circuit.**	A complete circuit and a source of potential difference.	☐
22	**State the equation linking charge flow, current and time.**	charge flow = current × time (Q = I t)	☐
23	**State the equation linking potential difference, current and resistance.**	potential difference = current × resistance (V = I R)	☐
24 ★	**State the condition for the current through a resistor to be directly proportional to the potential difference.**	Its temperature must stay constant.	☐

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